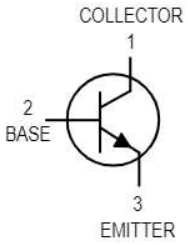
Amplifier Transistors
NPN Silicon

BC237,A,B,C
BC238B,C
BC239,C





CASE 29–04, STYLE 17
TO–92 (TO–226AA)

MAXIMUM RATINGS

Rating	Symbol	BC 237	BC 238	BC 239	Unit
Collector–Emitter Voltage	V_{CEO}	45	25	25	Vdc
Collector–Emitter Voltage	V_{CES}	50	30	30	Vdc
Emitter–Base Voltage	V_{EBO}	6.0	5.0	5.0	Vdc
Collector Current — Continuous	I_C	100			mAdc
Total Device Dissipation @ T_A = 25°C Derate above 25°C	P_D	350 2.8			mW mW/°C
Total Device Dissipation @ T_C = 25°C Derate above 25°C	P_D	1.0 8.0			Watts mW/°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	–55 to +150			°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	357	°C/W
Thermal Resistance, Junction to Case	$R_{\theta JC}$	125	°C/W

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Characteristic		Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS						
Collector–Emitter Breakdown Voltage (I_C = 2.0 mA, I_B = 0)	BC237 BC238 BC239	$V_{(BR)CEO}$	45 25 25	— — —	— — —	V
Emitter–Base Breakdown Voltage (I_E = 100 μA, I_C = 0)	BC237 BC238 BC239	$V_{(BR)EBO}$	6.0 5.0 5.0	— — —	— — —	V
Collector Cutoff Current		I_{CES}				
(V_{CE} = 30 V, V_{BE} = 0)	BC238 BC239		— —	0.2 0.2	15 15	nA
(V_{CE} = 50 V, V_{BE} = 0)	BC237		—	0.2	15	
(V_{CE} = 30 V, V_{BE} = 0) T_A = 125°C	BC238 BC239		— —	0.2 0.2	4.0 4.0	∝A
(V_{CE} = 50 V, V_{BE} = 0) T_A = 125°C	BC237		—	0.2	4.0	

REV 1

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ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted) (Continued)

Characteristic		Symbol	Min	Typ	Max	Unit
ON CHARACTERISTICS						
DC Current Gain		h_{FE}				—
(I_C = 10 ∝A, V_{CE} = 5.0 V)	BC237A		—	90	—	
	BC237B/238B		—	150	—	
	BC237C/238C/239C		—	270	—	
(I_C = 2.0 mA, V_{CE} = 5.0 V)	BC237		120	—	800	
	BC239		120	—	800	
	BC237A		120	170	220	
	BC237B/238B		200	290	460	
	BC237C/238C/239C		380	500	800	
(I_C = 100 mA, V_{CE} = 5.0 V)	BC237A		—	120	—	
	BC237B/238B		—	180	—	
	BC237C/238C/239C		—	300	—	
Collector–Emitter On Voltage		$V_{CE(sat)}$				V
(I_C = 10 mA, I_B = 0.5 mA)	BC237/BC238/BC239		—	0.07	0.2	
(I_C = 100 mA, I_B = 5.0 mA)	BC237/BC239		—	0.2	0.6	
	BC238				0.8	
Base–Emitter Saturation Voltage		$V_{BE(sat)}$				V
(I_C = 10 mA, I_B = 0.5 mA)			—	0.6	0.83	
(I_C = 100 mA, I_B = 5.0 mA)			—	—	1.05	
Base–Emitter On Voltage		$V_{BE(on)}$				V
(I_C = 100 ∝A, V_{CE} = 5.0 V)			—	0.5	—	
(I_C = 2.0 mA, V_{CE} = 5.0 V)			0.55	0.62	0.7	
(I_C = 100 mA, V_{CE} = 5.0 V)			—	0.83	—	
DYNAMIC CHARACTERISTICS						
Current–Gain — Bandwidth Product		f_T				MHz
(I_C = 0.5 mA, V_{CE} = 3.0 V, f = 100 MHz)	BC237		—	100	—	
	BC238		—	120	—	
	BC239		—	140	—	
(I_C = 10 mA, V_{CE} = 5.0 V, f = 100 MHz)	BC237		150	200	—	
	BC238		150	240	—	
	BC239		150	280	—	
Collector–Base Capacitance (V_{CB} = 10 V, I_C = 0, f = 1.0 MHz)		C_{obo}	—	—	4.5	pF
Emitter–Base Capacitance (V_{EB} = 0.5 V, I_C = 0, f = 1.0 MHz)		C_{ibo}	—	8.0	—	pF
Noise Figure		NF				dB
(I_C = 0.2 mA, V_{CE} = 5.0 V, R_S = 2.0 kΩ, f = 1.0 kHz)	BC239		—	2.0	4.0	
(I_C = 0.2 mA, V_{CE} = 5.0 V, R_S = 2.0 kΩ, f = 1.0 kHz, Δf = 200 Hz)	BC237		—	2.0	10	
	BC238		—	2.0	10	
	BC239		—	2.0	4.0	

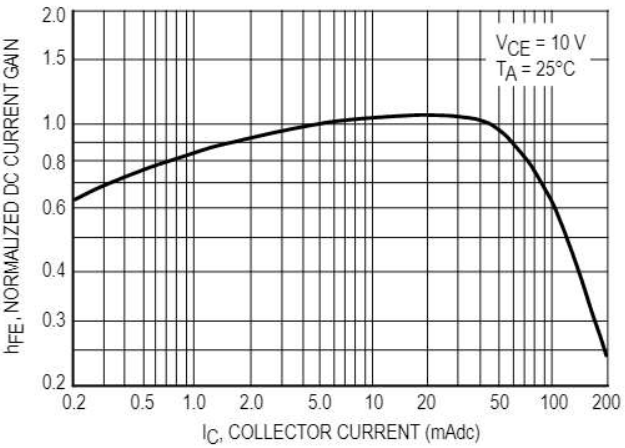


Figure 1. Normalized DC Current Gain

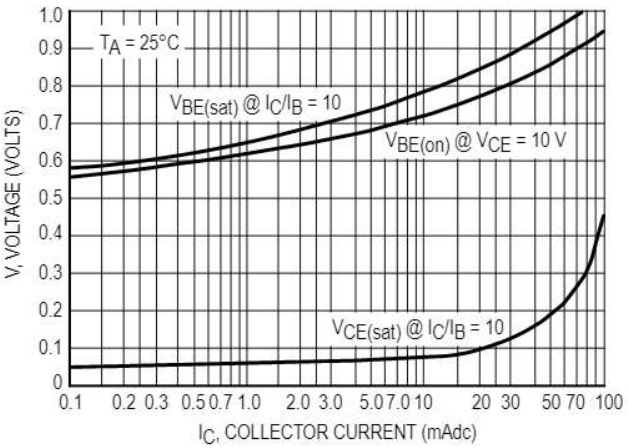


Figure 2. "Saturation" and "On" Voltages

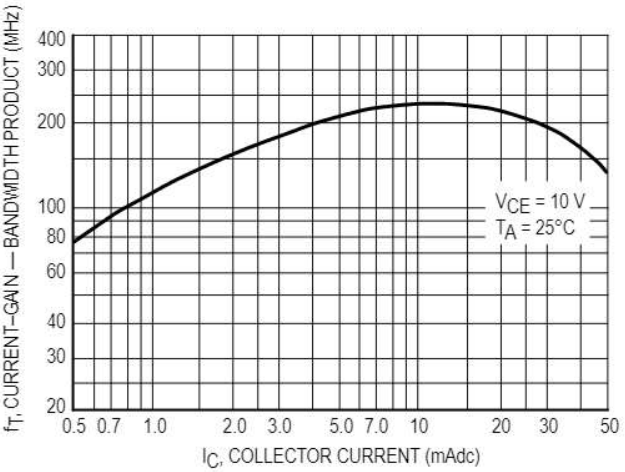


Figure 3. Current–Gain — Bandwidth Product

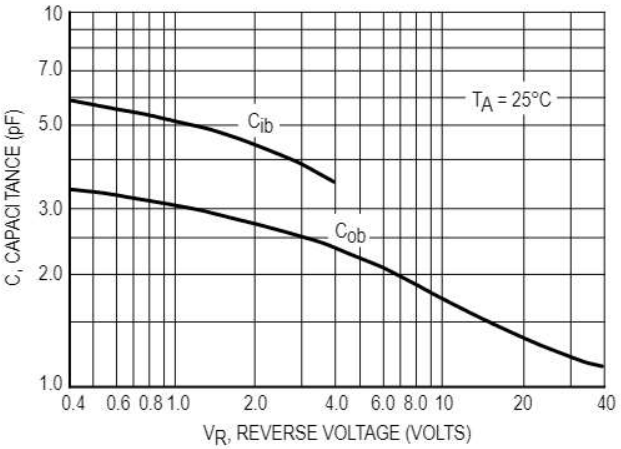


Figure 4. Capacitances

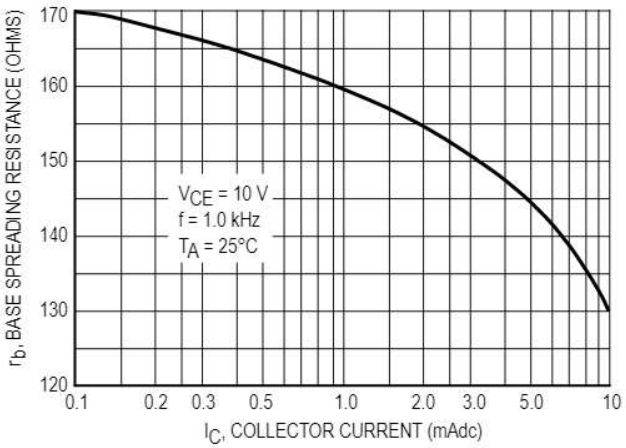
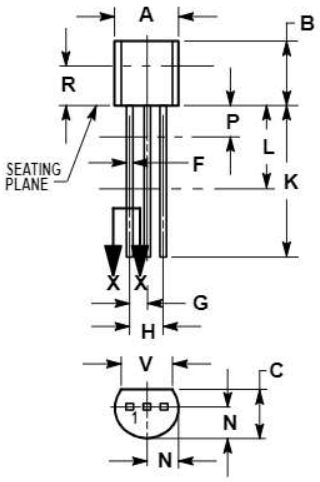
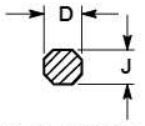


Figure 5. Base Spreading Resistance

PACKAGE DIMENSIONS





SECTION X–X

NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.
3. CONTOUR OF PACKAGE BEYOND DIMENSION R IS UNCONTROLLED.
4. DIMENSION F APPLIES BETWEEN P AND L. DIMENSION D AND J APPLY BETWEEN L AND K MINIMUM. LEAD DIMENSION IS UNCONTROLLED IN P AND BEYOND DIMENSION K MINIMUM.

	INCHES		MILLIMETERS	
DIM	MIN	MAX	MIN	MAX
A	0.175	0.205	4.45	5.20
B	0.170	0.210	4.32	5.33
C	0.125	0.165	3.18	4.19
D	0.016	0.022	0.41	0.55
F	0.016	0.019	0.41	0.48
G	0.045	0.055	1.15	1.39
H	0.095	0.105	2.42	2.66
J	0.015	0.020	0.39	0.50
K	0.500	----	12.70	----
L	0.250	----	6.35	----
N	0.080	0.105	2.04	2.66
P	----	0.100	----	2.54
R	0.115	----	2.93	----
V	0.135	----	3.43	----

CASE 029–04
(TO–226AA)
ISSUE AD

STYLE 17:
PIN 1. COLLECTOR
2. BASE
3. EMITTER

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Experiment (6)
BJT IV Characteristics

Objectives

- To study the IV curve of a BJT.
- To experimentally evaluate the DC gain parameter of a BJT.
- To study the transfer function of a BJT circuit.

Pre-lab Work

Analyze the DC circuit below. Assume the transistor is in active mode and find expressions for the collector current and the collector-emitter voltage.

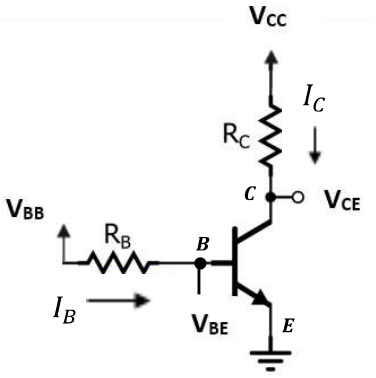


Figure 1: A BJT DC Circuit.

Solution

Equipment

Table 1. Equipment for experiment 8.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	Resistors 5 kΩ (1), 10 kΩ (1), 100 kΩ (1)			
3	BJT NPN transistor (1)	BC237	https://www.onsemi.com/pub/Collateral/2N3903-D.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): BJT IV Characteristics

1. Connect the circuit in figure 1 with R_C=10 kΩ and R_B=100 kΩ. Pay a special attention to how you connect the terminals of the BJT. Refer to the data sheet to identify the transistor terminals.

2. Set V_{BB} to a constant voltage of +10 V while connecting V_{CC} to a variable DC voltage source.
3. The IV characteristics (I_C versus V_{CE}) is obtained by slowly varying V_{CC} and measuring the voltage V_{CE} and V_{R_C} at each setting. Use table 2 to enter your results. Also record the measured values of V_{R_B} at each setting (to be used in part 2 of the experiment).

Part (2): BJT DC Gain Parameter (β)

1. The required data have been collected in part 1.

Part (3): BJT Transfer Function

1. Reconsider the circuit in figure 1.
2. Set V_{CC} to a constant voltage of +10 V while connecting V_{BB} to a variable DC voltage source.
3. The transfer function (V_{CE} versus V_{BB}) is obtained by slowly varying V_{BB} and measuring the voltage V_{CE} at each setting. Use table 3 to enter your results.
4. Repeat step 3 for another value of R_C, say 5 kΩ. Record your results in table 3.

Results and Analysis

Part (1): BJT IV Characteristics.

1. Calculate the current I_C values in table 2.
2. Use graph 1 to sketch the IV curve.

Table 2. IV Characteristics Measurements.

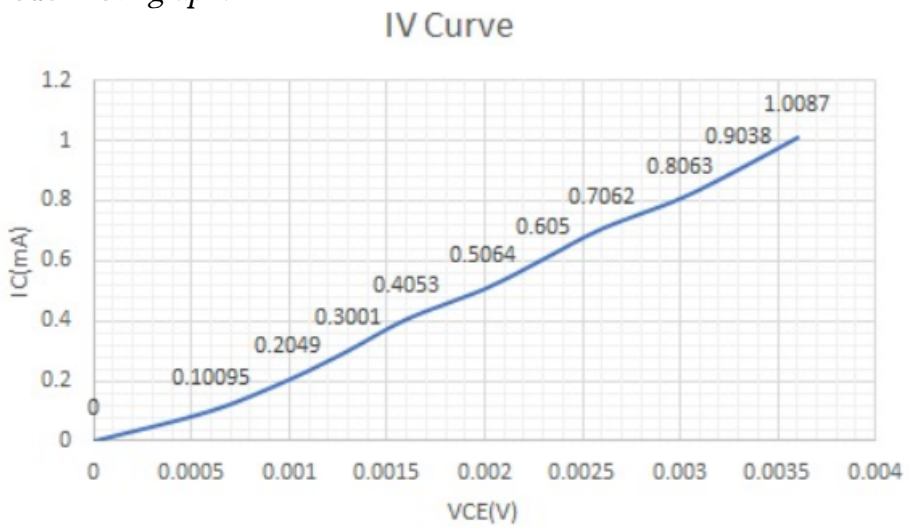
V_{CC} (V)	V_{CE} (V)	V_{R_C} (V)	I_C (mA)	V_{R_B} (V)	I_B (mA)	$\beta = I_C/I_B$
0	0	0	0	0	0	-
1	0.0006	1.0095	0.10095	9.2972	0.092972	1.085811
2	0.001	2.049	0.2049	9.2955	0.092955	2.204292
3	0.0013	3.0019	0.3001	9.2953	0.092953	3.228513
4	0.0016	4.0537	0.4053	9.2955	0.092955	4.360174
5	0.002	5.0642	0.5064	9.2556	0.092556	5.471282
6	0.0023	6.0502	0.605	9.2943	0.092943	6.509366
7	0.0026	7.0622	0.7062	9.2945	0.092945	7.598042
8	0.003	8.0634	0.8063	9.2997	0.092997	8.670172
9	0.0033	9.0384	0.9038	9.2536	0.092536	9.76701
10	0.0036	10.0872	1.0087	9.233	0.09233	10.92494

3. Comment on the IV curve that you obtained. Can you identify the transistor modes of operation? Show your answer on the curve.

Answer: *As a result of the IV curve we obtained the characteristic curve of a BJT, we can say that our transistor behaves as it is saturation region.*

4. Is the collector current constant in the active mode region of the IV curve? Find an approximate value of the Early voltage.

Answer: *Since we could not observe a result related to the active mode region in our graph, we could not make an inference about whether it is stable or not. Because our transistor is in saturation mode, we cannot comment or calculate the early voltage without switching to active mode. While the collector current should be constant according to the IV curve characteristic under normal conditions, we could not reach a conclusion regarding this because we could not observe the transition from saturation mode to active mode in our graph.*



Graph 1: Transistor IV Curve.

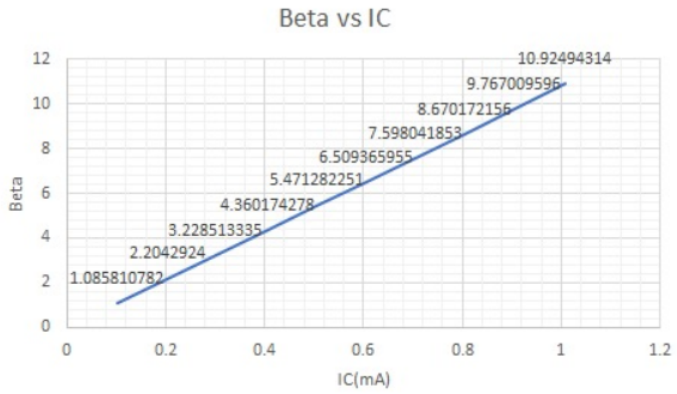
Part (2): BJT DC Gain Parameter (β)

1. Calculate the current I_B values in table 2. Also, calculate the DC gain parameter (β).
2. Why is the value of β not constant in the table? Explain in details.

Answer: *Beta value is not constant unless you keep the temperatureor current the same. Considering that our current values change during the experiment, we can explain why our beta is not constant.*

3. Use graph 2 to sketch β vs. I_C relation. Compare to the curve given in the transistor data sheet and give your comments.

Answer:



Graph 2: DC gain parameter.

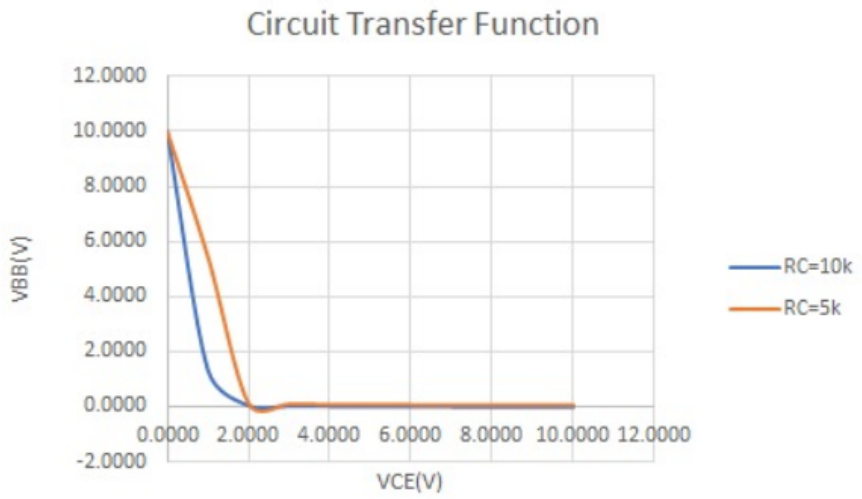
Part (3): BJT Transfer Function

1. Use graph 3 to sketch the two transfer functions using the data in table 3, for both R_C values.
2. Can you identify the transistor modes of operation? Show your answer on the curves.

Answer:

Table 3. Transfer Function Measurements.

	R_C= 10 kΩ	R_C= 5 kΩ
V_{BB} (V)	V_{CE} (V)	V_{CE} (V)
0.0000	10.0000	10.0000
1.0000	1.3649	5.4526
2.0000	0.0692	0.1035
3.0000	0.0492	0.0727
4.0000	0.0387	0.0587
5.0000	0.0319	0.0498
6.0000	0.0271	0.0435
7.0000	0.0235	0.0388
8.0000	0.0206	0.0349
9.0000	0.0183	0.0318
10.0000	0.0164	0.0292



Graph 3: Circuit transfer function.

3. Comment on the general shape of the transfer function. In particular, explain the main differences between the two curves, in relation to R_C value.

Answer: *As a result of the IV curve we obtained the characteristic curve of a BJT, we can say that our transistor behaves as it is saturation region.*

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name	Nurullah Said Tüğmen	Name	İlayda Kaya	Name		Name	
ID	150720823	ID	150720502	ID		ID	
Section	1						